

Bari Evidence Registry Memos for Annatto Sulphites and Selected Emulsifiers

Research framing and decision boundaries

These memos use only official regulatory sources, EFSA opinions and follow-ups, FDA/eCFR provisions, JECFA/WHO-FAO materials, Codex where useful, PubMed-indexed peer-reviewed papers, and official Israeli Ministry of Health materials. Where an exact hazard ratio or table value could not be verified directly from an accessible primary table in this session, it is explicitly left unquoted rather than inferred. That matters here, because Bari's evidence library should be conservative, implementation-ready, and resistant to overclaiming. ¹

Memo for annatto E160b

Executive conclusion

Annatto is a plant-derived food colour from the seeds of *Bixa orellana*; today, EU law separates it into **E160b(i) annatto bixin** and **E160b(ii) annatto norbixin** because the two colouring principles differ in chemistry, uses, and toxicology. EFSA's 2016 re-evaluation set ADIs of **6 mg/kg bw/day for bixin** and **0.3 mg/kg bw/day for norbixin**; EFSA's 2017 exposure statement found that **proposed** norbixin uses could exceed the norbixin ADI at high-percentile intake in toddlers and children, but EFSA's later 2019 follow-up indicated that exposure estimates for the assessed reported uses were below the ADIs, while concern remained mainly around overestimated or extended-use scenarios. JECFA is less restrictive numerically, with **0–12 mg/kg bw for bixin** and **0–0.6 mg/kg bw for norbixin expressed as norbixin**. In the US, FDA allows annatto extract as a colour additive exempt from certification, subject to GMP. In Israel, the Ministry of Health permitted-additives list allows annatto in relevant retail categories, with examples including **10 mg/kg in fats/spreads, 10 mg/kg in butter, 20 mg/kg in coatings/batters**, and permissions for certain cheeses and related products. The Bari bottom line is **likely-neutral**, not **functional**: the safety profile is broadly acceptable at authorised uses, but there is a documented **rare individual hypersensitivity signal** and a tighter exposure margin for norbixin than for bixin. ²

Evidence table

Source	Org/Journal	Year	URL/DOI/ PMID	Key points	Evidence strength	Registry implication
<i>The safety of annatto extracts (E 160b) as a food additive</i>	EFSA Journal	2016	DOI: 10.2903/ j.efsa. 2016.4544	EFSA re-evaluated annatto extracts, considered bixin- and norbixin-based extracts separately, set ADIs of 6 mg/kg bw/day for bixin and 0.3 mg/kg bw/day for norbixin, and reviewed rare hypersensitivity case evidence.	High	Core EU safety anchor; supports separate Bari handling of bixin/norbixin logic even if label still says E160b.
<i>Exposure assessment of annatto colouring principles bixin and norbixin (E 160b) when used as food additives</i>	EFSA Journal	2017	DOI: 10.2903/ j.efsa. 2017.4966; PMID 32625627	Bixin exposure did not exceed ADI; norbixin exposure exceeded ADI at high-percentile intake for toddlers and children in proposed-use scenarios.	High	Explains why annatto is not functional ; supports note on tighter exposure margin for norbixin.

Source	Org/Journal	Year	URL/DOI/ PMID	Key points	Evidence strength	Registry implication
<i>Safety of annatto E and the exposure to the annatto colouring principles bixin and norbixin (E 160b) when used as a food additive</i>	EFSA Journal	2019	DOI: 10.2903/ j.efsa. 2019.5626 ; PMID 32626252	Follow-up opinion reported exposure estimates below ADIs for assessed reported uses, while identifying that exceedance concerns were driven by certain assumptions and proposed/ extended uses.	High	Supports non-alarmist conclusion: authorised use can be acceptable, but norbixin margin is narrower.
Commission Regulation on annatto split and conditions of use	European Commission	2020	Commission Regulation (EU) 2020/771	Replaced E160b with E160b(i) annatto bixin and E160b(ii) annatto norbixin; set category-specific maximum levels and updated labelling terminology.	High	Strong basis for detection patterns and user-facing wording.
Annatto extract	FDA / eCFR	current	21 CFR 73.30	Annatto extract is exempt from certification and may be used for colouring foods generally under GMP, subject to standards-of-identity limits.	High	Supports US regulatory acceptance; no Bari safety escalation from US status alone.

Source	Org/Journal	Year	URL/DOI/PMID	Key points	Evidence strength	Registry implication
<i>Safety evaluation of certain food additives and contaminants</i>	WHO Food Additives Series	2007	WHO Food Additives Series 58	JECFA established ADI 0–12 mg/kg bw for bixin and a group ADI 0–0.6 mg/kg bw for norbixin and its sodium/potassium salts.	High	International anchor showing broad acceptance with different numerical ADIs from EFSA.
Israeli permitted-additives list	Israel Ministry of Health	2022	Official permitted additives list, fcs_list (22 Jun 2022)	Permits annatto in shelf-relevant categories including fats/spreads, butter, coatings/batters, and specified cheese categories with listed maxima.	High	Supports Israel-specific detection and category examples for Bari.

Risk interpretation

Established facts. Annatto is a colour additive of botanical origin, not an insect-derived colour like carmine. The modern EU regulatory frame distinguishes **bixin-based** and **norbixin-based** annatto because the two forms differ in extraction type, technological use, and toxicological reference values. That split is not cosmetic; it is the reason the same older shorthand label term “E160b” can hide materially different exposure limits. ³

Regulatory safety conclusion. The weight of official evidence is that annatto is **acceptable at authorised use levels**, but with a narrower safety margin for norbixin than for bixin. EFSA’s 2017 exposure statement is the key reason Bari should not treat annatto as a completely frictionless **functional** colour: EFSA found high-percentile norbixin exceedance under proposed-use assumptions in toddlers and children. EFSA’s subsequent 2019 opinion is equally important because it pulls the interpretation back from alarmism and indicates that exposures for the assessed reported uses were below ADIs. ⁴

Rare allergy and hypersensitivity evidence. The hypersensitivity signal for annatto is real, but it is best described as **rare and individual**, not population-level. EFSA’s 2016 review discusses case-level immediate reactions, including a report in which serum IgE recognised a 50-kDa annatto-extract protein thought to be a contaminant from the seed pericarp rather than the colouring principle itself. That makes the evidence

too thin for a broad consumer warning like “commonly allergenic,” but strong enough for an **optional contextual rare-reaction note** in Bari’s registry. ⁵

Dietary-preference and transparency relevance. Annatto is plant-derived, so it does **not** create the vegan/vegetarian or insect-origin transparency issue that applies to carmine. The real transparency issue is different: some labels still say only “E160b” or “annatto,” while the toxicology and EU conditions of use now distinguish bixin and norbixin. Bari should therefore treat annatto primarily as a **precision-labelling** issue, not a dietary-restriction issue. ⁶

Unsupported or exaggerated claims to avoid. Avoid saying that annatto is “dangerous,” “toxic,” or a “common allergen.” Avoid saying that “natural means safe” or that “natural colours are always harmless.” Avoid recommending routine avoidance. The evidence supports a narrower statement: *authorised colour, usually low concern, but with rare hypersensitivity reports and a tighter exposure margin for norbixin than for bixin.* ⁷

Bari tier recommendation

Recommended tier: likely-neutral

One-line justification: broad regulatory acceptance and no regulator conclusion of general-population harm at authorised food-use levels, but enough evidence of rare hypersensitivity and norbixin-specific exposure sensitivity to keep it out of functional. ⁸

What would move it up: durable exposure data showing comfortable margins for norbixin across current uses plus no further clinically relevant reaction signal.

What would move it down: replicated challenge-confirmed allergy evidence beyond isolated case reports, or new exposure evidence showing current-market norbixin intakes exceeding ADI in real populations.

Registry-ready entry

```
signal_id: additive_E160b_annatto_color
e_number: E160b
canonical: annatto_bixin_norbixin
ingredient_names:
  - E160b
  - E 160b
  - E-160b
  - E160b(i)
  - E160b(ii)
  - INS 160b
  - INS 160b(i)
  - INS 160b(ii)
  - annatto
  - annatto bixin
  - annatto norbixin
```

- bixin
- norbixin
- annatto extract
- annatto extracts
- Natural Orange 4
- CI 75120
- achiot

hebrew_patterns: "E\\s*[----]?\\s*160\\s*b(?:\\s*\\(?[ivxIVX]+\\)?)?|INS\\s*160\\s*b(?:\\s*\\(?[ivxIVX]+\\)?)?|צבע|נורבִּיקְסִין|בִּיקְסִין|אֲנָטוֹ|אֲנָטוֹן\\s*מַאֲכָל\\s*E\\s*160\\s*b"

english_patterns: "\\bE\\s*[----]?\\s*160\\s*b(?:\\s*\\(?[ivxIVX]+\\)?)?\\b|\\bINS\\s*160\\s*b(?:\\s*\\(?[ivxIVX]+\\)?)?\\b|\\bannat+t?o\\b|\\bannatto\\s+bixin\\b|\\bannatto\\s+norbixin\\b|\\bbixin\\b|\\bnorbixin\\b|\\bannatto\\s+extracts?\\b|\\bachiot\\b|\\bNatural\\s+Orange\\s*4\\b|\\bCI\\s*75120\\b"

additive_class: colorant_natural

proposed_d4_tier: likely-neutral

evidence_level: high

scoring_effect:

nutrition_score: "no_change; colour additive does not materially change nutrient quality"

additive_complexity: "mild_flag; added colour plus narrower norbixin margin than bixin"

dietary_preference: "none; plant-derived and generally compatible with vegan/vegetarian preferences"

allergy_caution: "optional_contextual_rare; rare hypersensitivity case reports exist"

confidence: 0.85

citations:

- "10.2903/j.efsa.2016.4544"
- "10.2903/j.efsa.2017.4966"
- "10.2903/j.efsa.2019.5626"
- "Commission Regulation (EU) 2020/771"
- "21 CFR 73.30"
- "WHO Food Additives Series 58"
- "Israel MoH permitted additives list 2022"

Detection patterns

Recommended regex set

- **Core E-number:** `\\bE\\s*[----]?\\s*160\\s*b(?:\\s*\\(?[ivxIVX]+\\)?)?\\b`
- **INS variants:** `\\bINS\\s*160\\s*b(?:\\s*\\(?[ivxIVX]+\\)?)?\\b`
- **English names:** `\\bannat+t?o\\b|\\bannatto\\s+bixin\\b|\\bannatto\\s+norbixin\\b|\\bbixin\\b|\\bnorbixin\\b|\\bannatto\\s+extracts?\\b|\\bachiot\\b|\\bNatural\\s+Orange\\s*4\\b|\\bCI\\s*75120\\b`
- **Hebrew names:** `צבע|נורבִּיקְסִין|בִּיקְסִין|אֲנָטוֹ|אֲנָטוֹן\\s*מַאֲכָל\\s*E\\s*160\\s*b`

Implementation note. Because Israeli OCR often drops parentheses or merges spacing, Bari should normalize hyphens, Hebrew apostrophes, and Latin/Hebrew punctuation before matching. The main practical risk is not missing “annatto”; it is missing old-style shorthand “E160b” when the exact extract type is not disclosed. ⁹

Scoring recommendation

Should E160b reduce the Bari nutrition score? No. It is a colour additive, not a nutritional quality marker. The evidence signal is about labelling precision and rare individual reactions, not nutrient risk. ¹⁰

Should it increase additive or UPF complexity? Yes, mildly. Reason: it is a cosmetic colour additive, and the bixin/norbixin distinction is more informative than most labels provide. ¹¹

Should it trigger a dietary-preference transparency flag? No dietary-restriction flag by default. It is plant-derived. A generic transparency note is optional if the product uses the non-specific term “E160b” rather than the more precise bixin/norbixin terminology. ⁶

Should it trigger an allergy/hypersensitivity caution? Yes, but only as optional contextual rare caution. Not a declared allergen. Do not surface as a red alert. ⁵

Safe consumer wording

- **English:** “Annatto is a plant-based colour. Regulators generally consider it acceptable at permitted levels. Rare sensitivity reactions have been reported in some individuals.”
- **Hebrew:** “אנאטו הוא צבע מאכל ממקור צמחי. ברמות השימוש המותרות הוא נחשב מקובל רגולטורית. דווחו מקרים.”
נדירים של רגישות אצל חלק מהאנשים

Memo for sulphur dioxide E220 and the sulphite family E220–E228

Executive conclusion

The sulphite family covers **sulphur dioxide and sulphites** used mainly as preservatives and antioxidants, labelled in Europe and Israel by their specific names or E-numbers and commonly interpreted as **SO₂ equivalents**. EFSA’s 2016 re-evaluation set a **temporary group ADI of 0.7 mg/kg bw/day as SO₂ equivalents**; EFSA’s **2022 follow-up** withdrew that temporary group ADI because new biological and toxicological data were insufficient and shifted to a **Margin-of-Exposure** approach instead. A later EFSA exposure update used an **MOE target of at least 80** for dietary exposure assessment. In parallel, the clinically important point did not change: sulphites are a **declared allergen/intolerance trigger** because they can provoke bronchospasm and related reactions in sensitive people, especially a subset of people with asthma, at real food-use exposures. The EU allergen-labelling threshold is **above 10 mg/kg or 10 mg/L expressed as SO₂**; FDA uses **10 ppm**; Israel’s Ministry of Health also states **10 mg/kg or 10 mg/L** as the threshold for sulphite disclosure and the official permitted-additives list allows E220–E228 in categories including shelf-relevant **dried fruits at 2000 mg/kg** and some wines at **200–250 mg/L**. Bari should classify the family under **confirmed-negative** for disclosure purposes, not because it is broadly unsafe for everyone, but because it has a **confirmed, clinically meaningful harm pathway in a susceptible subgroup at ordinary food-use exposures** and a strong regulatory labelling basis. ¹²

Evidence table

Source	Org/Journal	Year	URL/DOI/ PMID	Key points	Evidence strength	Registry implication
<i>Follow-up of the re-evaluation of sulfur dioxide (E220) and sulfites (E221–228) as food additives</i>	EFSA Journal	2022	DOI: 10.2903/j.efsa.2022.7594 ; PMID 36440381	Confirms 2016 temporary group ADI of 0.7 mg/kg bw/day as SO ₂ equivalents, states data gaps remained, and withdraws the temporary group ADI.	High	Core EU anchor for Bari: do not describe sulphites as “cleared with a stable ADI”; the current EFSA frame is more cautious.
<i>Update of the dietary exposure to sulfur dioxide (E220), ... sulfites (E228) as food additives</i>	EFSA Supporting Publications	2025	10.2903/sp.efsa.2025.EN-9754	Uses MOE framework and states dietary exposure should achieve an MOE of at least 80.	High	Supports exposure-sensitive framing and warns against blanket “always safe” language.
Sulphur dioxide and sulphites allergen labelling basis	European Union	current	Regulation (EU) No 1169/2011, Annex II	Requires disclosure of sulphur dioxide and sulphites above 10 mg/kg or 10 mg/L expressed as SO ₂ .	High	Strong basis for consumer-facing Bari disclosure flag.

Source	Org/Journal	Year	URL/DOI/ PMID	Key points	Evidence strength	Registry implication
Sulfites in standardized food	FDA / eCFR	current	21 CFR 130.9	FDA defines a detectable amount as 10 ppm or more sulfite in the finished food and requires declaration.	High	Supports US regulatory convergence on label relevance rather than broad toxicity rhetoric.
FDA sulfite labelling notice	Federal Register / FDA	2022	FDA docket / notice on sulfites analysis	FDA reiterates that sulfites at 10 ppm or more must be labelled and acknowledges allergic-type responses in sensitive individuals.	High	Reinforces declared- caution logic.
<i>Clinical effects of sulphite additives</i>	Clinical & Experimental Allergy	2009	PMID 19775253	Reviews respiratory, cutaneous and other adverse reactions; clinically relevant reactions occur mainly in sensitive individuals, especially some people with asthma.	Medium- High	Best clinical grounding for Bari's "sensitive subgroup" wording.

Source	Org/Journal	Year	URL/DOI/ PMID	Key points	Evidence strength	Registry implication
<i>Adverse reactions to the sulphite additives</i>	Gastroenterology & Hepatology from Bed to Bench	2012	PMCID available via PMC	Reviews mechanisms and notes that both IgE and non-IgE mechanisms have been proposed, with non-IgE pathways important in many cases.	Medium	Supports non-simplistic mechanism note; avoid overselling classic IgE allergy.
Israeli permitted-additives list	Israel Ministry of Health	2022	Official permitted additives list, fcs_list (22 Jun 2022)	Allows E220–E228 in multiple categories; relevant examples include dried fruits and wines with listed maxima.	High	Israel-specific category anchor for Bari.
Israeli sulphite allergen guidance	Israel Ministry of Health	current	Official allergen guidance page	States sulphites are reported when present above 10 mg/kg or 10 mg/L .	High	Directly supports Israel-facing consumer flag.

Source	Org/Journal	Year	URL/DOI/ PMID	Key points	Evidence strength	Registry implication
Sulfites evaluation	WHO / JECFA	classic	WHO Food Additives Series 42	JECFA historically retained a group ADI for sulphites expressed as SO ₂ and identified major contributors such as dried fruit and non- alcoholic beverages.	High	International reference; supports category- level relevance.

Risk interpretation

Established facts. Sulphites are not a narrow single additive but a regulatory family: E220, E221, E222, E223, E224, E226, E227, and E228. In practice, risk-management and labelling are handled on an SO₂-equivalent basis, which is why threshold wording and maximum levels are often expressed “as sulphur dioxide.” ¹³

Regulatory safety conclusion. For the general population, the correct regulatory summary is **not** “sulphites are banned” and **not** “sulphites are fully cleared without issue.” EFSA moved from a temporary group ADI to an MOE approach because the submitted follow-up data did not adequately close the toxicology gaps. That means Bari should avoid language that sounds more certain than EFSA’s present stance. At the same time, regulatory systems in the EU, US, and Israel still permit sulphites in many food categories with maximum levels and mandatory disclosure thresholds. ¹⁴

Rare allergy versus confirmed subgroup risk. This is the key judgement call. Sulphites are **not** like annatto, where the reaction evidence is mainly isolated case reporting. For sulphites, there is a long-standing, well-recognised clinical pathway of **sulphite-sensitive asthma and other hypersensitivity-type reactions**, especially in susceptible individuals. Mechanistically, these reactions are not reliably reducible to one pathway; reviews describe both IgE and non-IgE possibilities, with non-IgE and irritant/bronchoconstrictive pathways often emphasized. This is why Bari should treat sulphites as a **declared caution family**, not merely a weak observational signal. ¹⁵

Population-level exposure versus high-intake extremes. The official material available here is strongest on the regulatory shift and the need for MOE-based assessment, and weaker—in publicly accessible snippet form—on clean category-by-category consumer MOEs for every common product. The safest interpretation is therefore narrower: **high-use categories such as dried fruits and some wines are the practical relevance zones**, and sensitive people can react at ordinary food-use exposures even where general-population toxicology would not justify broad alarm. Bari should not claim that typical Israeli exposure for

all consumers clearly “breaches EFSA MOE” across the board; the evidence supports a more limited statement about **susceptible individuals** and **high-use categories**. ¹⁶

Dietary-preference relevance. Sulphites are not a vegan/vegetarian issue and do not themselves create a kosher/halal source-opacity problem. Their relevance is functional and clinical, not identity-based. ¹⁷

Unsupported or exaggerated claims to avoid. Avoid “sulphites are toxic for everyone,” “sulphites cause cancer,” and “everyone should avoid sulphites.” Also avoid describing sulphites as a classic food allergy in all cases; many reactions are better framed as **sulphite sensitivity/hypersensitivity**, especially respiratory reactions in susceptible people. ¹⁸

Bari tier recommendation

Recommended tier: confirmed-negative

One-line justification: not because they are broadly hazardous to everyone, but because there is a **confirmed clinically relevant adverse pathway at real food-use exposures in a susceptible subgroup**, plus explicit regulatory disclosure thresholds in the EU, US, and Israel. ¹⁹

What would move it up: evidence that the declared-allergen/hypersensitivity concern is clinically negligible at real-world labelled exposures, which is not how current regulation treats sulphites.

What would move it down further: more direct evidence that ordinary consumer intakes in common categories systematically compress EFSA’s MOE for the general population, not just for sensitive individuals or high consumers.

Registry-ready entry

```
signal_id: additive_E220_sulphites_family
e_number: E220
canonical: sulphur_dioxide_sulphites_family
ingredient_names:
  - E220
  - E221
  - E222
  - E223
  - E224
  - E226
  - E227
  - E228
  - sulfur dioxide
  - sulphur dioxide
  - sulfites
  - sulphites
  - sulfur dioxide-sulfites
  - sulfur dioxide and sulfites
```

- ```
hebrew_patterns: "E\\s*[---]?\\s*22(?0|1|2|3|4|6|7|8)\\b|INS\\s*22(?0|1|2|3|4|6|7|8)\\b|תהג ופרית|סולפ יט(ים?:)?\\s*דו[---]?חמצנ ית|דו[---]?תחמוצת\\s*ג ופרית|מטה[---]?ביס וולפיט|אשלג \\s*ס וולפיט(י?:)?ם|ביס וולפיט|מטה[---]?ביס וולפיט|מטה[---]?ביס וולפיט\\s*"

```

```
additive_class: preservative_sulphite
```

evidence\_level: high

```
nutrition_score: "no_change; preservative status does not directly change
nutrient quality"
```

```
dietary_preference: "none"
```

confidence: 0.93

- "10.2903/j.efsa.2022.7594"
- "10.2903/sp.efsa.2025.EN-9754"
- "Regulation (EU) No 1169/2011 Annex II"
- "21 CFR 130.9"
- "PMID 19775253"
- "WHO Food Additives Series 42"
- "Israel MoH permitted additives list 2022"
- "Israel MoH sulphite disclosure guidance"

## Detection patterns

### Recommended regex set

- **Family E-numbers:** `\bE\s*[---]?s*22(?:0|1|2|3|4|6|7|8)\b`
- **INS variants:** `\bINS\s*22(?:0|1|2|3|4|6|7|8)\b`
- **English names:**  
`\bsulphur\s+dioxide\b|\bsulfur\s+dioxide\b|\bsulphites?\b|\bsulfites?\b|`  
`\bsodium\s+sul(?:f|ph)ite\b|\bsodium\s+bisul(?:f|ph)ite\b|`  
`\bsodium\s+metabisul(?:f|ph)ite\b|\bpotassium\s+metabisul(?:f|ph)ite\b|`  
`\bpotassium\s+bisul(?:f|ph)ite\b|\bcalcium\s+sul(?:f|ph)ite\b|`  
`\bcalcium\s+bisul(?:f|ph)ite\b`
- **Hebrew names:**  
`מטה[---]ביסולפיט|מטהס*דין[---]חמצנית|דון[---]תחמוצתס*גופרית`  
`מטה[---]ביסולפיט|מטהס*דין[---]חמצנית|דון[---]תחמוצתס*גופרית`

**Implementation note.** Bari should treat **E220** as the parent canonical entry and absorb E221–E228 into the same family rule so that products labelled only by chemistry name still trigger the same disclosure logic.

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## Scoring recommendation

**Should sulphites reduce the Bari nutrition score? No.** The issue is not nutrient quality; it is preservative use and clinically relevant sensitivity/disclosure. <sup>21</sup>

**Should they increase additive or UPF complexity? Yes.** At least a **moderate additive-complexity flag** is justified because this is a preservation family with clear high-use categories and disclosure rules. <sup>22</sup>

**Should they trigger a dietary-preference flag? No.** Not primarily a vegan/vegetarian/religious-source issue. <sup>23</sup>

**Should they trigger an allergy caution? Yes — definitely.** This is the clearest Bari action point. Use declared-caution wording, not scare wording. <sup>19</sup>

### Safe consumer wording

- **English:** “Sulphites are preservatives used in foods such as dried fruit, wine, and some processed products. Most people tolerate them, but sensitive individuals — especially some people with asthma — can react, which is why they are specifically disclosed on labels.”
- **Hebrew:** “סולפיטים הם חומרים משמרים שנמצאים למשל בפירות יבשים, יין ומוצרים מעובדים מסוימים. רוב האנשים סובלים אותם היטב, אך אצל אנשים רגישים — במיוחד חלק מהאנשים עם אסתמה — הם עלולים לעורר תגובה, ולכן יש להם חובת סימון ברורה על גבי התווית.”

# Memo for E435 polysorbate 60, E475 polyglycerol esters, and E477 propylene glycol esters

## Executive conclusion

These three additives are all emulsifiers used mainly for texture, aeration, and stability in bakery and dessert-type foods, but they do **not** sit on equally strong evidence footing. **E435 polysorbate 60** belongs to the polysorbate family; EFSA re-evaluated polysorbates collectively and retained a **group ADI of 25 mg/kg bw/day**, JECFA retains **0–25 mg/kg bw/day**, FDA permits it in specific foods including cakes and icings with category limits, and Israel’s permitted-additives list gives shelf-relevant permissions including **5 g/kg in emulsified sauces (except mayonnaise)**. **E475 polyglycerol esters of fatty acids** has a more reassuring official toxicology profile, with EFSA concluding there was **no need for a numerical ADI**, though JECFA continues to use **0–25 mg/kg bw/day**; Israel’s list includes examples such as **0.5 g/kg in coffee whitener** and other higher-use categories. **E477 propane-1,2-diol esters of fatty acids** was re-evaluated by EFSA with an ADI of **25 mg/kg bw/day expressed as propane-1,2-diol**, corresponding to **80 mg/kg bw/day as E477**, and EFSA found **no safety concern at reported uses**; JECFA retains **0–25 mg/kg bw/day**, FDA allows it at GMP, and Israel’s list includes **10 g/kg in fat emulsions intended for baking**. The human-evidence question is where the memo matters for Bari: BMJ 2023 did **not** study E435 or E477 individually for CVD because consumption prevalence was too low; PLOS 2024 did **not** identify robust direct cancer signals for E435 or E477, while E475 showed a **non-robust** breast-cancer association that disappeared across sensitivity analyses. An official Lancet Diabetes & Endocrinology snippet indicates that **E435, E475, and E477 were among emulsifiers associated with higher type 2 diabetes risk** in NutriNet-Sant , but the exact hazard ratios could not be verified directly from the accessible primary tables in this session and should therefore **not** be quoted in Bari copy yet. The right Bari answer is not to auto-inherit E433/E471’s exact status blindly: **E475 merits the strongest individual contestation, E435 a lower-confidence contested rating, and E477 a contested rating only with low confidence and strict wording discipline.** <sup>24</sup>

## Evidence table

| Source                                                                                                                                    | Org/Journal  | Year | URL/DOI/PMID                                                | Key points                                                                                                                            | Evidence strength | Registry implication                 |
|-------------------------------------------------------------------------------------------------------------------------------------------|--------------|------|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|-------------------|--------------------------------------|
| <i>Scientific Opinion on the re-evaluation of polyoxyethylene sorbitan monolaurate... monostearate (E435)... and related polysorbates</i> | EFSA Journal | 2015 | DOI: <b>10.2903/j.efsa.2015.4152</b> ; PMID <b>42109790</b> | Acute toxicity very low; no concern re genotoxicity, carcinogenicity, or developmental toxicity; group ADI retained for polysorbates. | High              | Official toxicology anchor for E435. |

| Source                                                                               | Org/Journal          | Year    | URL/DOI/PMID                                                | Key points                                                                                                    | Evidence strength | Registry implication                                                                            |
|--------------------------------------------------------------------------------------|----------------------|---------|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-------------------|-------------------------------------------------------------------------------------------------|
| Polysorbate 60 monograph                                                             | FAO JECFA Monographs | 2014    | <b>FAO JECFA Monographs 16</b>                              | JECFA specification lists Polysorbate 60 / INS 435 and states ADI <b>0–25 mg/kg bw.</b>                       | High              | International anchor for E435.                                                                  |
| Polysorbate 60                                                                       | FDA / eCFR           | current | <b>21 CFR 172.836</b>                                       | FDA permits polysorbate 60 in specified food categories with quantitative limits, including cakes and icings. | High              | Supports US regulatory acceptance; not a nutrition-score issue.                                 |
| <i>Re-evaluation of polyglycerol esters of fatty acids (E475) as a food additive</i> | EFSA Journal         | 2017    | DOI: <b>10.2903/j.efsa.2017.5089</b> ; PMID <b>32625376</b> | EFSA concluded there was <b>no need for a numerical ADI</b> at reported uses/use levels.                      | High              | Official toxicology anchor for E475; distinct from class-level observational signals.           |
| <i>Follow-up of the re-evaluation of polyglycerol esters of fatty acids (E475)</i>   | EFSA Journal         | 2022    | DOI: <b>10.2903/j.efsa.2022.7308</b> ; PMID <b>35600272</b> | Follow-up focused on specifications/ data gaps.                                                               | High              | Supports caution about impurity/ specification questions but not a direct food-dose harm claim. |
| Polyglycerol esters specification                                                    | FAO JECFA Monographs | 2025    | <b>FAO JECFA Monographs 34</b>                              | Reaffirmed JECFA ADI <b>0–25 mg/kg bw</b> for INS 475.                                                        | High              | International anchor for E475.                                                                  |

| Source                                                                                   | Org/Journal  | Year    | URL/DOI/PMID                                                | Key points                                                                                                                                                                      | Evidence strength | Registry implication                                           |
|------------------------------------------------------------------------------------------|--------------|---------|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|----------------------------------------------------------------|
| Polyglycerol esters of fatty acids                                                       | FDA / eCFR   | current | <b>21 CFR 172.854</b>                                       | FDA permits use in food under prescribed conditions.                                                                                                                            | High              | Supports regulatory acceptance.                                |
| <i>Re-evaluation of propane-1,2-diol esters of fatty acids (E477) as a food additive</i> | EFSA Journal | 2018    | DOI: <b>10.2903/j.efsa.2018.5497</b> ; PMID <b>32625782</b> | EFSA confirmed ADI of <b>25 mg/kg bw/day expressed as propane-1,2-diol</b> , equivalent to <b>80 mg/kg bw/day as E477</b> , and found no safety concern at reported use levels. | High              | Official toxicology anchor for E477.                           |
| Propylene glycol esters monograph                                                        | FAO / JECFA  | 1997    | <b>FNP 52 Add. 5 / Combined Compendium entry</b>            | JECFA specification states ADI <b>0–25 mg/kg bw</b> for INS 477.                                                                                                                | High              | International anchor for E477.                                 |
| Propylene glycol mono- and diesters of fats and fatty acids                              | FDA / eCFR   | current | <b>21 CFR 172.856</b>                                       | FDA allows use in food at amounts reasonably required for effect.                                                                                                               | High              | Supports regulatory acceptance.                                |
| <i>Food additive emulsifiers and risk of cardiovascular disease</i>                      | BMJ          | 2023    | DOI: <b>10.1136/bmj-2023-076058</b>                         | E435 and E477 were among emulsifiers consumed by <b>&lt;5%</b> of participants and therefore were <b>not studied individually</b> for CVD risk.                                 | Medium-High       | Important negative qualifier; avoid inventing class spillover. |

| Source                                                                                              | Org/Journal                     | Year | URL/DOI/PMID                                                    | Key points                                                                                                                                                                                                                                   | Evidence strength | Registry implication                                                                   |
|-----------------------------------------------------------------------------------------------------|---------------------------------|------|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|----------------------------------------------------------------------------------------|
| <i>Food additive emulsifiers and cancer risk</i>                                                    | PLOS Medicine                   | 2024 | DOI: <b>10.1371/journal.pmed.1004338</b> ; PMID <b>38349899</b> | E475 showed a breast-cancer association in main model ( <b>HR 1.50; 95% CI 1.05–2.15</b> ) but it was <b>not robust</b> across sensitivity analyses; E435 and E477 were listed as consumed additives but not among robust positive findings. | Medium-High       | Best human-evidence signal for E475; weaker for E435/E477.                             |
| <i>Food additive emulsifiers and the risk of type 2 diabetes</i>                                    | Lancet Diabetes & Endocrinology | 2024 | DOI/Article ID: <b>S2213-8587(24)00086-X</b>                    | Official snippet indicates E435, E475, and E477 were among emulsifiers associated with higher type 2 diabetes risk; exact HRs were not directly verifiable from accessible primary tables in this session.                                   | Medium            | Supports <b>contested</b> framing, but Bari should not quote exact HRs until verified. |
| <i>Dietary emulsifiers impact the mouse gut microbiota promoting colitis and metabolic syndrome</i> | Nature                          | 2015 | DOI: <b>10.1038/nature14232</b> ; PMID <b>25731162</b>          | Mechanistic evidence of gut-barrier/microbiota effects is strongest for <b>polysorbate 80 (E433)</b> and CMC, not these three additives directly.                                                                                            | Medium            | Justifies caution against over-extending sibling evidence.                             |

| Source                                                                                                 | Org/Journal               | Year | URL/DOI/PMID                                                     | Key points                                                                                                      | Evidence strength | Registry implication                                                               |
|--------------------------------------------------------------------------------------------------------|---------------------------|------|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-------------------|------------------------------------------------------------------------------------|
| <i>Influence of surface-active food additives on the integrity and absorption of intestinal mucosa</i> | Toxicology                | 1984 | PMID <b>6094320</b>                                              | Older mucosal-cell study found both polysorbate 60 and polysorbate 80 could affect intestinal-cell integrity.   | Low-Medium        | Gives E435 some mechanistic plausibility, but far weaker than modern P80 evidence. |
| Israeli permitted-additives list                                                                       | Israel Ministry of Health | 2022 | Official permitted additives list, <b>fcs_list</b> (22 Jun 2022) | Provides Israel-specific category permissions and maxima for E435, E475, and E477 in shelf-relevant categories. | High              | Essential for Bari implementation in Israel.                                       |

## Risk interpretation

**Established facts.** All three are authorised emulsifiers with ordinary food-technology functions: aeration, stabilization, texture, and prevention of phase separation. None is a nutrient issue. None is a declared allergen family. The main Bari question is whether long-term human-health signals justify a contested interpretation despite reassuring classical toxicology. <sup>25</sup>

**Regulatory safety conclusion.** The official toxicology picture remains relatively reassuring. EFSA and JECFA support use of E435 and E477 within established ADIs, and EFSA found no need for a numerical ADI for E475 at reported uses. FDA also permits all three under food-additive provisions. So if Bari rated only from classical toxicology, these would land closer to neutral than to negative. <sup>25</sup>

**What the cohort evidence actually supports.** This is where overreach is easy. BMJ 2023 explicitly did **not** analyse E435 or E477 individually for CVD because too few participants consumed them. PLOS 2024 did not produce a robust direct cancer signal for E435 or E477; for E475, the main-model breast-cancer signal was **not robust** across sensitivity analyses. The 2024 Lancet Diabetes paper appears, from official snippet text, to have included E435, E475, and E477 among emulsifiers associated with increased type 2 diabetes risk, but the exact HRs were not directly retrievable from the accessible primary tables here. Bari therefore has enough to justify a **contested** label on evidence librarianship grounds, but **not** enough to publish precise risk magnitudes for consumers without a secondary verification pass. <sup>26</sup>

**Does the E433 polysorbate-80 mechanism extend to E435 polysorbate-60?** Only **partially and inferentially**. The strongest modern barrier/microbiome evidence is on **E433/P80**, not E435. E435 shares the polyoxyethylene-sorbitan-ester scaffold, and older intestinal-mucosa work did include polysorbate 60 alongside polysorbate 80, which gives some plausibility to a family-level concern. But Bari should not

phrase E435 as if the P80 evidence has been directly proven for it. “Related polysorbate with lower-confidence class spillover” is more accurate than “same risk as E433.” <sup>27</sup>

**Dietary-preference and source-opacity relevance.** For **E475 and E477**, the fatty-acid source can be plant or animal unless the manufacturer specifies otherwise; that creates a genuine **vegan/vegetarian/kosher/halal transparency issue**. The same source-opacity concern can also arise for E435 because the stearic-acid component can be from different origins, though labels usually do not surface that detail. Bari should therefore treat these not as prohibited by default, but as **source-opaque emulsifiers** where dietary-preference certainty is limited from the label alone. <sup>28</sup>

**Unsupported or exaggerated claims to avoid.** Avoid “all emulsifiers damage the gut,” “E435 is basically the same as E433,” “human harm is proven,” and “these should lower the nutrition score.” Also avoid publishing exact hazard ratios for E435/E475/E477 from the diabetes paper until the original supplementary tables are directly checked. <sup>29</sup>

## **Bari tier recommendation**

**E435 polysorbate 60:** contested

**Why:** human signal appears to exist in the diabetes cohort and there is some family-level mechanistic plausibility, but direct evidence remains materially weaker than for E433. <sup>30</sup>

**What would move it up:** failure to replicate in human cohorts or direct negative human intervention data.

**What would move it down further:** direct verified cohort HRs plus modern mechanistic work showing P60-specific barrier or microbiome disruption at realistic exposures.

**E475 polyglycerol esters of fatty acids:** contested

**Why:** it has the strongest individual human signal of the trio, with a non-robust cancer association and apparent inclusion in the diabetes-signal set, even though official toxicology remains relatively reassuring. <sup>31</sup>

**What would move it up:** replicated null results in future cohorts or direct negative intervention studies.

**What would move it down further:** a replicated individual association across diabetes/CVD/cancer cohorts that survives sensitivity analyses.

**E477 propylene glycol esters of fatty acids:** contested **with low confidence**

**Why:** official toxicology is reassuring and CVD/cancer evidence is weak, but the Lancet diabetes signal appears to include E477 among the associated emulsifiers. Because the exact effect size was not directly verifiable here, Bari should keep wording especially conservative. <sup>32</sup>

**What would move it up:** a verified null or very weak diabetes estimate in the primary table plus absence of replication.

**What would move it down further:** verified individual HRs and replication in other cohorts or direct human mechanistic evidence.

## Registry-ready entries

```
signal_id: additive_E435_polysorbate60
e_number: E435
canonical: polysorbate_60
ingredient_names:
 - E435
 - E 435
 - E-435
 - INS 435
 - polysorbate 60
 - polyoxyethylene (20) sorbitan monostearate
 - polyoxyethylene sorbitan monostearate
hebrew_patterns: "E\\s*[----]?\\s*435\\b|INS\\s*435\\b|פוליסורבט\\s*60|
מונוסטארט\\s*סורביטן\\s*פוליסורבט-60|פוליוקסיאתילן"
english_patterns: "\\bE\\s*[----]?\\s*435\\b|\\bINS\\s*435\\b|\\bpolyorbate\\
\\s*60\\b|\\bpolyoxyethylene\\s*\\(?20\\)?\\s*sorbitan\\s*monostearate\\b|\\
\\bpolyoxyethylene\\s+sorbitan\\s+monostearate\\b"
additive_class: emulsifier_polysorbate
proposed_d4_tier: contested
evidence_level: moderate
scoring_effect:
 nutrition_score: "no_change; emulsifier use is not a nutrient-quality marker"
 additive_complexity: "mild_flag; cosmetic/texture emulsifier with low-
confidence contested signal"
 dietary_preference: "contextual_source_opaque; fatty-acid source may not be
label-clear"
 allergy_caution: "none"
confidence: 0.64
citations:
 - "10.2903/j.efsa.2015.4152"
 - "FAO JECFA Monographs 16"
 - "21 CFR 172.836"
 - "10.1136/bmj-2023-076058"
 - "S2213-8587(24)00086-X"
 - "10.1038/nature14232"
 - "PMID 6094320"
```

```
signal_id: additive_E475_polyglycerol_esters
e_number: E475
canonical: polyglycerol_esters_fatty_acids
ingredient_names:
 - E475
 - E 475
 - E-475
 - INS 475
```

- polyglycerol esters of fatty acids
- polyglycerol fatty acid esters
- glycerin fatty acid esters
- PGFE
- PGE

hebrew\_patterns: "E\\s\*[-]??\\s\*475\\b|INS\\s\*475\\b|אסטרם\\s\*של\\s\*חומצות\\s\*שומן\\s\*פוליגליצרול\\s\*חומצות\\s\*פוליגליצרול\\s\*שומן\\s\*אסטרם\\s\*של\\s\*חומצות\\s\*שומן"

english\_patterns: "\\bE\\s\*[-]??\\s\*475\\b|\\bINS\\s\*475\\b|\\bpolyglycerol\\s+esters\\s+of\\s+fatty\\s+acids\\b|\\bpolyglycerol\\s+fatty\\s+acid\\s+esters\\b|\\bglycerin\\s+fatty\\s+acid\\s+esters\\b|\\bPGFE\\b|\\bPGE\\b"

additive\_class: emulsifier\_fatty\_acid\_ester

proposed\_d4\_tier: contested

evidence\_level: moderate

scoring\_effect:

- nutrition\_score: "no\_change; emulsifier status does not directly alter nutrient quality"
- additive\_complexity: "mild\_flag; emulsifier with observational signal stronger than for most peers"
- dietary\_preference: "contextual\_source\_opaque; fatty acids may be plant- or animal-derived"
- allergy\_caution: "none"

confidence: 0.71

citations:

- "10.2903/j.efsa.2017.5089"
- "10.2903/j.efsa.2022.7308"
- "FAO JECFA Monographs 34"
- "21 CFR 172.854"
- "10.1371/journal.pmed.1004338"
- "S2213-8587(24)00086-X"

signal\_id: additive\_E477\_propylene\_glycol\_esters

e\_number: E477

canonical: propylene\_glycol\_esters\_fatty\_acids

ingredient\_names:

- E477
- E 477
- E-477
- INS 477
- propylene glycol esters of fatty acids
- propane-1,2-diol esters of fatty acids
- propylene glycol mono- and diesters of fats and fatty acids
- PGMS

hebrew\_patterns: "E\\s\*[-]??\\s\*477\\b|INS\\s\*477\\b|אסטרם\\s\*של\\s\*פרופילן\\s\*חומצות\\s\*גליקול\\s\*חומצות\\s\*פרופילן\\s\*אסטרם\\s\*של\\s\*חומצות\\s\*שומן\\s\*גליקול\\s\*חומצות\\s\*שומן"

english\_patterns: "\\bE\\s\*[-]??\\s\*477\\b|\\bINS\\s\*477\\b|\\bpropylene\\s+glycol\\s+esters\\s+of\\s+fatty\\s+acids\\b|\\bpropane-1,2-diol\\s+esters\\s+of\\s+fatty\\s+acids\\b|\\bpropylene\\s+glycol\\s+mono-\\s+and\\s+diesters\\s+of\\s+fats\\s+and\\s+fatty\\s+acids\\b|\\bPGMS\\b"

```

\s+glycol\s+esters\s+of\s+fatty\s+acids\b|\bpropane-?1,2-?diol\s+esters\
\s+of\s+fatty\s+acids\b|\bpropylene\s+glycol\s+mono-?\s*and\s+diesters\
\s+of\s+fats\s+and\s+fatty\s+acids\b|\bPGMS\b"
additive_class: emulsifier_fatty_acid_ester
proposed_d4_tier: contested
evidence_level: low_to_moderate
scoring_effect:
 nutrition_score: "no_change; not a nutrient-quality marker"
 additive_complexity: "mild_flag; emissifier with low-confidence contested
signal and source opacity"
 dietary_preference: "contextual_source_opaque; fatty acid source may be
plant- or animal-derived"
 allergy_caution: "none"
confidence: 0.58
citations:
 - "10.2903/j.efsa.2018.5497"
 - "FNP 52 Add. 5"
 - "21 CFR 172.856"
 - "10.1136/bmj-2023-076058"
 - "S2213-8587(24)00086-X"

```

## Detection patterns

### E435

- \bE\s\*[\---]?s\*435\b
- \bINS\s\*435\b
- \bpolyorbate\s\*60\b
- \bpolyoxyethylene\s\*?20?s\*sorbitan\s\*monostearate\b
- Hebrew: מונוסטארטס\סורביטן\פוליוקסיאתילן\s\*60\פוליסורבט

### E475

- \bE\s\*[\---]?s\*475\b
- \bINS\s\*475\b
- \bpolyglycerol\s+esters\s+of\s+fatty\s+acids\b
- \bpolyglycerol\s+fatty\s+acid\s+esters\b
- Hebrew: שומן\סחומצות\s\*פולילגליצרול\s\*של\s\*שומן\אסטרים\סחומצות\s\*של\s\*אסטרים\פולילגליצרול

### E477

- \bE\s\*[\---]?s\*477\b
- \bINS\s\*477\b
- \bpropylene\s+glycol\s+esters\s+of\s+fatty\s+acids\b
- \bpropane-?1,2-?diol\s+esters\s+of\s+fatty\s+acids\b

- `\bpropylene\s+glycol\s+mono-?`  
`\s*and\s+diesters\s+of\s+fats\s+and\s+fatty\s+acids\b`
- Hebrew:  
`שומן\s+וחומצות\s*גליקול\s*פרופילן\s*של\s*שומן\s*אסטרים\s*חומצות\s*של\s*אסטרים\s*גליקול\s*פרופילן`

**OCR normalization tip.** Normalize hyphens, brackets, and spacing before matching. These additives often appear in bilingual labels with inconsistent spacing, for example “E 475”, “E-475”, or English chemistry names without E-numbers. <sup>33</sup>

## Scoring recommendation

**Should these reduce the Bari nutrition score? No for all three.** The evidence question concerns additive function and possible long-term observational signals, not nutrient quality. <sup>34</sup>

**Should they increase additive or UPF complexity? Yes, mildly for each, and more if multiple emulsifiers stack in one product.** These are texture/cosmetic emulsifiers with some observational and class-level concern, but they do not justify a nutrition penalty. <sup>35</sup>

**Should they trigger a dietary-preference transparency flag? Yes, contextual source-opacity flag for E475 and E477, and optional contextual flag for E435.** The label usually does not reveal whether the fatty-acid component is plant- or animal-derived. <sup>28</sup>

**Should they trigger an allergy caution? No default allergy caution.** The relevant concern is not a declared-allergen pathway. <sup>34</sup>

## Safe consumer wording

- **English:** “These are emulsifiers used to improve texture and stability. Food-safety authorities permit them at regulated levels. Some observational research has raised questions about certain emulsifiers, but the evidence is not settled and does not justify a nutrition penalty.”
- **Hebrew:** “אלה אמולסיפיקטורים שנועדו לשפר מרקם ויציבות. רשויות בטיחות מזון מתירות את השימוש בהם ברמות מוסדרות. קיימים מחקרים תצפיתיים שמעלים שאלות לגבי חלק מהאמולסיפיקטורים, אך הראיות אינן סופיות ואינן מצדיקות הורדת ציון תזונתי.”

## Operational handoff

### Assessment

The registry decisions are implementable now, but one claim should stay out of consumer-facing copy until a direct table check is done: the **exact hazard ratios** for **E435/E475/E477** in the 2024 diabetes paper. The current evidence is already strong enough to implement Bari tiers, detection rules, and wording without waiting for that extra verification step. <sup>36</sup>

### Owner

**Primary owner:** Research Analyst

## Why this owner

This is an evidence-library update with regulatory synthesis, wording boundaries, and structured registry output. Implementation should then pass to Product, Nutrition, and QA, but the evidence lock should sit with Research first.

## Copy-paste prompt

### Research Analyst

"Finalize three Bari registry records exactly as drafted here. Keep annatto as **likely-neutral**, sulphites as **confirmed-negative**, and E435/E475/E477 as separate entries with contested logic and conservative wording. Do not add exact E435/E475/E477 diabetes HRs unless you verify them directly from the primary supplementary tables."

### Head of Product

"Approve scoring policy: no nutrition-score reduction for annatto, sulphites, E435, E475, or E477. Apply additive-complexity flags instead. For sulphites, require an explicit declared-caution surface. For annatto, keep caution optional and rare. For E435/E475/E477, use mild complexity and contextual transparency rather than warning-style UX."

### Chief Nutrition Officer

"Approve the consumer language in English and Hebrew. Keep all copy non-alarmist, avoid 'dangerous' or 'toxic,' and preserve the distinction between general-population safety and subgroup sensitivity."

### Frontend Architect

"Implement the regex patterns and canonical mapping exactly as specified. Treat E220 as parent for the sulphite family. Normalize OCR spacing, hyphens, Hebrew punctuation, and INS variants before matching."

### QA & Audit Lead

"Create regression tests for: E160b/E160b(i)/E160b(ii); Hebrew sulphite chemistry names and E220–E228; E435/E475/E477 with spacing and OCR variants. Add negative tests to ensure polysorbate 80 evidence is not accidentally copied onto E435 consumer copy as a proven direct effect."

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1 2 3 6 9 11 <https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX%3A32020R0771>  
<https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX%3A32020R0771>

4 7 8 <https://efsa.onlinelibrary.wiley.com/doi/10.2903/j.efsa.2.17.4966>  
<https://efsa.onlinelibrary.wiley.com/doi/10.2903/j.efsa.2.17.4966>

- 5 [https://www.researchgate.net/publication/319536977\\_The\\_safety\\_of\\_annatto\\_extracts\\_E\\_160b\\_as\\_a\\_food\\_additive](https://www.researchgate.net/publication/319536977_The_safety_of_annatto_extracts_E_160b_as_a_food_additive)  
[https://www.researchgate.net/publication/319536977\\_The\\_safety\\_of\\_annatto\\_extracts\\_E\\_160b\\_as\\_a\\_food\\_additive](https://www.researchgate.net/publication/319536977_The_safety_of_annatto_extracts_E_160b_as_a_food_additive)
- 10 <https://pubmed.ncbi.nlm.nih.gov/32626252/>  
<https://pubmed.ncbi.nlm.nih.gov/32626252/>
- 12 13 21 <https://pubmed.ncbi.nlm.nih.gov/36440381/>  
<https://pubmed.ncbi.nlm.nih.gov/36440381/>
- 14 <https://efsa.onlinelibrary.wiley.com/doi/10.2903/j.efsa.2022.7594>  
<https://efsa.onlinelibrary.wiley.com/doi/10.2903/j.efsa.2022.7594>
- 15 18 19 <https://pubmed.ncbi.nlm.nih.gov/19775253/>  
<https://pubmed.ncbi.nlm.nih.gov/19775253/>
- 16 <https://www.bfr.bund.de/cm/349/sulphur-dioxide-and-sulphites-efsa-re-evaluation-identifies-health-risks-at-high-consumption-data-situation-still-incomplete.pdf>  
<https://www.bfr.bund.de/cm/349/sulphur-dioxide-and-sulphites-efsa-re-evaluation-identifies-health-risks-at-high-consumption-data-situation-still-incomplete.pdf>
- 17 20 <https://www.ecfr.gov/current/title-21/chapter-I/subchapter-B/part-130/subpart-A/section-130.9>  
<https://www.ecfr.gov/current/title-21/chapter-I/subchapter-B/part-130/subpart-A/section-130.9>
- 22 [https://www.gov.il/BlobFolder/reports/fcs\\_list/he/files\\_publications\\_food\\_food-addictive-libarary\\_fcs\\_list.pdf](https://www.gov.il/BlobFolder/reports/fcs_list/he/files_publications_food_food-addictive-libarary_fcs_list.pdf)  
[https://www.gov.il/BlobFolder/reports/fcs\\_list/he/files\\_publications\\_food\\_food-addictive-libarary\\_fcs\\_list.pdf](https://www.gov.il/BlobFolder/reports/fcs_list/he/files_publications_food_food-addictive-libarary_fcs_list.pdf)
- 23 <https://www.gov.il/he/pages/adv-food-labeling>  
<https://www.gov.il/he/pages/adv-food-labeling>
- 24 25 34 <https://www.cabidigitallibrary.org/doi/pdf/10.5555/20153289354>  
<https://www.cabidigitallibrary.org/doi/pdf/10.5555/20153289354>
- 26 29 [https://www.researchgate.net/publication/373721064\\_Food\\_additive\\_emulsifiers\\_and\\_risk\\_of\\_cardiovascular\\_disease\\_in\\_the\\_NutriNet-Sante\\_cohort\\_prospective\\_cohort\\_study](https://www.researchgate.net/publication/373721064_Food_additive_emulsifiers_and_risk_of_cardiovascular_disease_in_the_NutriNet-Sante_cohort_prospective_cohort_study)  
[https://www.researchgate.net/publication/373721064\\_Food\\_additive\\_emulsifiers\\_and\\_risk\\_of\\_cardiovascular\\_disease\\_in\\_the\\_NutriNet-Sante\\_cohort\\_prospective\\_cohort\\_study](https://www.researchgate.net/publication/373721064_Food_additive_emulsifiers_and_risk_of_cardiovascular_disease_in_the_NutriNet-Sante_cohort_prospective_cohort_study)
- 27 <https://pubmed.ncbi.nlm.nih.gov/25731162/>  
<https://pubmed.ncbi.nlm.nih.gov/25731162/>
- 28 <https://www.ecfr.gov/current/title-21/chapter-I/subchapter-B/part-172/subpart-I/section-172.854>  
<https://www.ecfr.gov/current/title-21/chapter-I/subchapter-B/part-172/subpart-I/section-172.854>
- 30 32 35 36 <https://www.thelancet.com/journals/landia/article/PIIS2213-8587%2824%2900086-X/fulltext>  
<https://www.thelancet.com/journals/landia/article/PIIS2213-8587%2824%2900086-X/fulltext>
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<https://journals.plos.org/plosmedicine/article?id=10.1371%2Fjournal.pmed.1004338>
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